Step 3 — Scaling to 3 Tools With a Hard Step Limit and Failure Recovery — Solution

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Story — Step 3: a one-tool demo never shows what breaks for real. Real tools call networks that fail, and real questions leave the model choosing between tools. This step adds two tools — one real API, one that fails on purpose — and the Worker from Step 2 handles them **without any change to its loop**: `run_tool()` already turns failures into observations, and the step limit already exists. **If not:** the first real network failure would crash the app — or you'd rewrite the loop now, instead of just adding tools.

After this step the Worker answers live weather, company policy and conversions, and recovers from a tool failure. Files not shown here don't change.

The new tools

Changed from Step 2: `exceptions.py` adds Doc02's two HTTP errors, used by the HTTP client below.

```
# src/exceptions.py
# The project's named errors: MissingConfigError (Doc01),
# TransientHTTPError and PermanentHTTPError (Doc02), and
# MaxIterationsExceeded (Doc07), which carries the partial step log.

class MissingConfigError(Exception):
    """Raised when a required setting is missing from the environment."""
    pass

class TransientHTTPError(Exception):
    """Raised when a retryable failure (timeout, 5xx, 429) exhausts all
    attempts."""
    pass

class PermanentHTTPError(Exception):
    """Raised immediately for a non-retryable 4xx, with the response body
    attached."""

    def __init__(self, status_code: int, body: str) -> None:
```

```
        self.status_code = status_code
        self.body = body
        super().__init__(f"HTTP {status_code}: {body[:200]}")

class MaxIterationsExceeded(Exception):
    def __init__(self, message: str, steps: list[dict]) -> None:
        self.steps = steps
        super().__init__(message)
```

Story — `utils/http_client.py`: Doc02's Build Task client (Intermediate), with one change: the logger is fetched inside the function, like every other file here, so `LOG_LEVEL` from `.env` is loaded first. **If not:** the weather tool would call `requests` directly, with no timeout and no retry.

```
# src/utils/http_client.py
# request_with_retry(): Doc02's HTTP client — a timeout on every call,
# retries with backoff for temporary errors, and a named error for each
# kind of failure.
import json
import random
import time
from typing import Any

import requests

from src.exceptions import PermanentHTTPError, TransientHTTPError
from src.utils.logger import get_logger

DEFAULT_TIMEOUT = (3, 10)

def compute_backoff_delay(attempt: int) -> float:
    return (2 ** attempt) + random.uniform(0, 1)

def request_with_retry(
    method: str,
    url: str,
    max_attempts: int = 5,
    **kwargs: Any,
) -> dict:
    logger = get_logger(__name__)
    for attempt in range(max_attempts):
        logger.debug(f"attempt {attempt}: {method} {url}")
        try:
            response = requests.request(
                method, url, timeout=DEFAULT_TIMEOUT, **kwargs
            )
        except (
```

```
            requests.exceptions.Timeout,
            requests.exceptions.ConnectionError,
        ) as e:
            if attempt == max_attempts - 1:
                logger.error(f"gave up after {max_attempts} attempts: {e}")
                raise TransientHTTPError(str(e)) from e
            time.sleep(compute_backoff_delay(attempt))
            continue

        if response.status_code < 300:
            try:
                return response.json()
            except json.JSONDecodeError as e:
                raise TransientHTTPError(f"bad JSON in response: {e}") from e

        if response.status_code == 429 or response.status_code // 100 == 5:
            if attempt == max_attempts - 1:
                logger.error(f"gave up, last status {response.status_code}")
                raise TransientHTTPError(
                    f"status {response.status_code} "
                    f"after {max_attempts} attempts"
                )
            time.sleep(compute_backoff_delay(attempt))
            continue

        logger.error(f"permanent failure: status {response.status_code}")
        raise PermanentHTTPError(response.status_code, response.text)
```

Story — `tools/docs_search.py`: a small, local policy "search" with known answers. It **fails on its first call on purpose**, using Doc07's `FlakyToolState`, so every run proves the Worker recovers. **If not:** failure recovery could only be tested by waiting for a real outage.

```
# src/tools/docs_search.py
# The policy search tool: search_docs(query) looks up the company's refund,
# shipping and warranty rules. It fails on its first call on purpose, so
# the Worker's recovery can be tested.
from langchain_core.tools import tool

COMPANY_DOCS = {
    "refund": "Refunds are accepted within 30 days of purchase, "
    "with the original receipt.",
    "shipping": "Standard shipping takes 3-5 working days; express "
    "shipping takes 1 working day.",
    "warranty": "Every product has a 1-year warranty against "
    "manufacturing faults.",
}
```

```
class FlakyToolState:
    def __init__(self) -> None:
        self.call_count = 0

search_state = FlakyToolState()

@tool
def search_docs(query: str) -> str:
    """Search the company's policy documents: refunds, shipping and
    warranty. Use it only for company policy questions."""
    search_state.call_count = search_state.call_count + 1
    if search_state.call_count == 1:
        raise ConnectionError("search index temporarily unavailable")

    lowered = query.lower()
    for topic in COMPANY_DOCS:
        if topic in lowered:
            return COMPANY_DOCS[topic]
    return "No matching document found."
```

Solution — a weather tool on Doc02's client, and a Worker that only gains tools

Story — `tools/weather.py`: Doc06's real-API tool, calling Doc02's `request_with_retry("GET", ...)`, so it gets timeouts and retries. The description says when *not* to use it — with 3 tools, the description is how the model chooses. **If not:** the model could reach for the weather tool on a policy question, and one slow response could hang the run.

```
# src/tools/weather.py
# The weather tool: get_weather(city) looks up the city's coordinates and
# calls the Open-Meteo API through request_with_retry().
from langchain_core.tools import tool

from src.utils.http_client import request_with_retry

WEATHER_URL = "https://api.open-meteo.com/v1/forecast"

CITY_COORDINATES = {
    "lahore": {"latitude": 31.55, "longitude": 74.34},
    "karachi": {"latitude": 24.86, "longitude": 67.01},
    "islamabad": {"latitude": 33.68, "longitude": 73.05},
    "london": {"latitude": 51.51, "longitude": -0.13},
    "paris": {"latitude": 48.85, "longitude": 2.35},
```

```
}

def build_weather_params(city: str) -> dict | None:
    coords = CITY_COORDINATES.get(city.lower())
    if coords is None:
        return None
    return {
        "latitude": coords["latitude"],
        "longitude": coords["longitude"],
        "current_weather": True,
    }

@tool
def get_weather(city: str) -> str:
    """Get the current temperature in Celsius for a city. Use it only for
    live weather questions — never for company policy or conversions."""
    params = build_weather_params(city)
    if params is None:
        return f"No coordinates on file for {city}."

    data = request_with_retry("GET", WEATHER_URL, params=params)
    temperature = data["current_weather"]["temperature"]
    return f"Currently {temperature}°C in {city}"
```

Changed from Step 2: `prompts/worker.txt` is rewritten for 3 tools and the retry rule.

Story — `prompts/worker.txt`: says which kinds of questions need tools, that a failed tool may be tried once more, and that the answer must come from what the tools returned. **If not:** the model might answer a policy question from memory, or give up after the first error.

`src/prompts/worker.txt`

```
You are a research assistant. Use your tools for anything live, exact or
company-specific: weather, temperature conversions and company policy.
Answer general knowledge directly, without tools. If a tool returns an
error, you may try it again once. Base your final answer only on what the
tools returned, and say so if a tool could not give you the answer.
```

Changed from Step 2: in `worker.py`, **only** two imports and the `ALL_TOOLS` list change. The loop, `find_tool()`, `run_tool()`, the step log and the limit are exactly as you wrote them in Step 2.

```
# src/agents/worker.py
# The Worker agent: run_worker() is the ReAct loop — ask the model, run the
# tool it asks for, send the result back — with a step log and a hard
# step limit. ALL_TOOLS lists the tools it may use.
from dataclasses import dataclass, field
```

```
from langchain_core.tools import BaseTool
from langchain_openai import ChatOpenAI

from src.exceptions import MaxIterationsExceeded
from src.tools.converter import celsius_to_fahrenheit
# new in Step 3: the 2 new tools
from src.tools.docs_search import search_docs
from src.tools.weather import get_weather
from src.utils.logger import get_logger
from src.utils.prompts import load_prompt

# changed in Step 3: 2 tools added to the list
ALL_TOOLS = [get_weather, celsius_to_fahrenheit, search_docs]

@dataclass
class AgentResult:
    final_answer: str
    steps: list[dict] = field(default_factory=list)

def find_tool(tools: list[BaseTool], name: str) -> BaseTool | None:
    for candidate in tools:
        if candidate.name == name:
            return candidate
    return None

def run_tool(
    call: dict, tools: list[BaseTool]
) -> tuple[str | None, str | None]:
    tool = find_tool(tools, call["name"])
    if tool is None:
        return None, f"unknown tool: {call['name']}"
    try:
        result = tool.invoke(call["args"])
        return str(result), None
    except Exception as e:
        return None, str(e)

def run_worker(
    model: ChatOpenAI, task: str, max_iterations: int
) -> AgentResult:
    logger = get_logger(__name__)
    model_with_tools = model.bind_tools(ALL_TOOLS)
    messages = [
        {"role": "system", "content": load_prompt("worker")},
        {"role": "user", "content": task},
    ]
```

```
    steps: list[dict] = []

    for step in range(max_iterations):
        response = model_with_tools.invoke(messages)
        messages.append(response)

        if not response.tool_calls:
            return AgentResult(final_answer=response.content, steps=steps)

        for call in response.tool_calls:
            result, error = run_tool(call, ALL_TOOLS)
            if error is None:
                observation = result
                logger.info("step %s: %s -> %s", step, call["name"], result)
            else:
                observation = f"Error: {error}"
                logger.warning(
                    "step %s: %s failed -> %s", step, call["name"], error
                )
            messages.append({
                "role": "tool",
                "tool_call_id": call["id"],
                "content": observation,
            })
            steps.append({
                "step": step,
                "tool": call["name"],
                "arguments": call["args"],
                "result": result,
                "error": error,
            })

    message = f"No answer after {max_iterations} steps"
    raise MaxIterationsExceeded(message, steps)
```

Changed from Step 2: `tests/unit/test_tools.py` adds cases 3, 4 and 5 — three new functions and their imports.

Story — `tests/unit/test_tools.py`: cases 3 and 4 check the weather tool's coordinate lookup without the network (a known city, and an unknown one); case 5 proves the search tool fails once and then works. **If not:** a broken tool would only show up as a strange agent answer.

```
# tests/unit/test_tools.py
# Checks the tools. Each case prints OK or WRONG.
# No API calls, no API key, no network needed.
# Run from the repo root: python -m tests.unit.test_tools
```

```
from pydantic import ValidationError

from src.tools.converter import celsius_to_fahrenheit
# new in Step 3
from src.tools.docs_search import search_docs, search_state
from src.tools.weather import build_weather_params


def test_converter() -> None:
    result = "OK"
    if celsius_to_fahrenheit.invoke({"celsius": 18}) != 64.4:
        result = "WRONG"
    print(f"case 1 (18°C is 64.4°F)           -> {result}")


def test_converter_rejects_bad_arguments() -> None:
    result = "WRONG"
    try:
        celsius_to_fahrenheit.invoke({"celsius": "hot"})
    except ValidationError:
        result = "OK"
    print(f"case 2 (celsius='hot' refused)    -> {result}")


# new in Step 3
def test_weather_params_known_city() -> None:
    params = build_weather_params("Lahore")
    result = "OK"
    if params is None or params["latitude"] != 31.55:
        result = "WRONG"
    print(f"case 3 (Lahore coordinates)       -> {result}")


# new in Step 3
def test_weather_params_unknown_city() -> None:
    result = "OK"
    if build_weather_params("Atlantis") is not None:
        result = "WRONG"
    print(f"case 4 (unknown city gives None)  -> {result}")


# new in Step 3
def test_search_fails_then_recovers() -> None:
    search_state.call_count = 0
    result = "WRONG"
    try:
        search_docs.invoke({"query": "refund policy"})
```

```
    except ConnectionError:
        result = "OK"
    second = search_docs.invoke({"query": "refund policy"})
    if "30 days" not in second:
        result = "WRONG"
    print(f"case 5 (fails once, then works)   -> {result}")


if __name__ == "__main__":
    test_converter()
    test_converter_rejects_bad_arguments()
    # new in Step 3
    test_weather_params_known_city()
    test_weather_params_unknown_city()
    test_search_fails_then_recovers()
```

Changed from Step 2: the integration test adds case 5 (a weather question uses `get_weather`) and case 6 (the Worker recovers from the failing search).

```
# tests/integration/test_live_agents.py
# Checks the app against the real OpenAI API (and Open-Meteo) — each step
# adds cases. Each case prints OK or WRONG.
# Needs a valid OPENAI_API_KEY in .env.
# Run from the repo root: python -m tests.integration.test_live_agents
from langchain_openai import ChatOpenAI

from src.agents.worker import AgentResult, run_worker
from src.chains.direct_answer import build_direct_chain
from src.config import load_config
from src.exceptions import MaxIterationsExceeded
from src.models.llm import get_model
# new in Step 3
from src.tools.docs_search import search_state


def test_direct_answer(model: ChatOpenAI) -> None:
    chain = build_direct_chain(model)
    answer = chain.invoke({"question": "What is the capital of France?"})
    result = "OK"
    if "Paris" not in answer:
        result = "WRONG"
    print(f"case 1 (direct answer)  -> {result}: {answer}")


def tools_used(outcome: AgentResult) -> list[str]:
    names = []
```

```
    for step in outcome.steps:
        names.append(step["tool"])
    return names


def test_conversion_uses_tool(model: ChatOpenAI) -> None:
    task = "What is 25 degrees Celsius in Fahrenheit?"
    used = tools_used(run_worker(model, task, 5))
    result = "OK"
    if "celsius_to_fahrenheit" not in used:
        result = "WRONG"
    print(f"case 2 (conversion)     -> {result}: {used}")


def test_no_tool_needed(model: ChatOpenAI) -> None:
    outcome = run_worker(model, "What is the capital of France?", 5)
    result = "OK"
    if len(outcome.steps) != 0:
        result = "WRONG"
    print(f"case 3 (no tool needed) -> {result}: {outcome.final_answer}")


def test_step_limit(model: ChatOpenAI) -> None:
    task = "What is 25 degrees Celsius in Fahrenheit?"
    result = "WRONG"
    try:
        run_worker(model, task, 1)
    except MaxIterationsExceeded:
        result = "OK"
    print(f"case 4 (step limit)     -> {result}")


# new in Step 3
def test_weather_uses_tool(model: ChatOpenAI) -> None:
    task = "What's the weather in Lahore right now?"
    used = tools_used(run_worker(model, task, 5))
    result = "OK"
    if "get_weather" not in used:
        result = "WRONG"
    print(f"case 5 (weather task)   -> {result}: {used}")


# new in Step 3
def test_tool_failure_recovery(model: ChatOpenAI) -> None:
    search_state.call_count = 0
    outcome = run_worker(model, "What is your refund policy?", 5)
    failed = 0
```

```
    succeeded = 0
    for step in outcome.steps:
        if step["error"] is None:
            succeeded = succeeded + 1
        else:
            failed = failed + 1
    result = "OK"
    if failed == 0 or succeeded == 0:
        result = "WRONG"
    print(f"case 6 (recovery)       -> {result}: {failed} failed, "
          f"{succeeded} succeeded")


if __name__ == "__main__":
    config = load_config()
    model = get_model(config)
    test_direct_answer(model)
    test_conversion_uses_tool(model)
    test_no_tool_needed(model)
    test_step_limit(model)
    # new in Step 3
    test_weather_uses_tool(model)
    test_tool_failure_recovery(model)
```

`main.py` doesn't change in this step — the new tools simply appear in the printed step log.

Run everything:

```
python -m tests.unit.test_config
python -m tests.unit.test_tools
python -m tests.unit.test_agents
python -m tests.integration.test_live_agents
python main.py
```

Expected output of the new `test_tools` cases:

```
case 3 (Lahore coordinates)       -> OK
case 4 (unknown city gives None)  -> OK
case 5 (fails once, then works)   -> OK
```

Expected output of `test_live_agents` — the Step 2 cases, then (one `WARNING` log line appears for the failed search):

```
case 5 (weather task)   -> OK: ['get_weather']
```

```
case 6 (recovery)       -> OK: 1 failed, 1 succeeded
```

Expected app session:

```
You: What is your refund policy?
... WARNING step 0: search_docs failed -> search index temporarily
unavailable
  step 0: search_docs({'query': 'refund policy'})
          -> Error: search index temporarily unavailable
  step 1: search_docs({'query': 'refund policy'})
          -> Refunds are accepted within 30 days of purchase, with the
             original receipt.
Answer: Refunds are accepted within 30 days of purchase, with the
original receipt.
```

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